Digital Electronics & Finite State Machine Design

Course: Digital Electronics | Semester 3 | Contributor: Rohan Verma

K-map minimization, flip-flop conversions, synchronous counters, and race-around conditions.

Combinational Logic Optimization

* Karnaugh maps for 4 and 5 variables with don’t care conditions.
* Quine-McCluskey tabular reduction method for multiple output circuits.
* Decoders, encoders, multiplexer trees, and full-adder design with NAND gates.

Sequential Circuits & State Machines

* JK Flip-Flop race-around elimination via Master-Slave and edge-triggering.
* Excitation tables for SR, JK, D, and T flip-flops.
* Mealy vs Moore state machine models and state diagram synthesis.

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